

Evoking familiarity without recollection: Modeling the déjà vu experience



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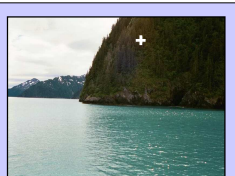
Introduction

We attempted to model a feeling of false positive familiarity similar to the déjà vu experience. After shallow exposure to pictures of naturalistic scenes (searching for an embedded cross), participants returned 1 (Experiment 1) or 3 (Experiment 2) weeks later to evaluate whether they had actually visited each pictured setting (new plus old scenes). Mundane and unique campus scenes from an unfamiliar campus (Duke or SMU) were shown either 0, 1 or 2 times in the first session. The mean visit rating increased with any prior exposure, but 1 versus 2 exposures did not differ. Participants gave higher visit ratings to mundane than unique scenes, and this did not vary as a function of exposure frequency or delay. About half of each sample reported "yes" to question "did you have a déjà vu response to any of the photos?"

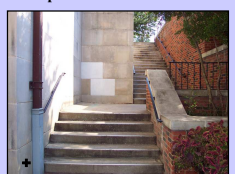
Experiment 1 Method

Participants. 36 SMU and 42 Duke Students

Session 1. 48 non-home campus photos (SMU photos for Duke students; Duke photos for SMU students) were presented once (24 photos) or twice (24 photos), plus 144 filler photos. Half of the campus photos were mundane and half were unique. Each photo had a cross (+) in one of its four quadrants and participants pressed one of four computer keys to indicate the cross quadrant.



Example of Session 1 Filler Photo



Example of Session 1 Critical Photo SMU-Mundane

Session 2 (1 week later). 144 campus shots (72 SMU; 72 Duke) plus 48 community photos (24 Dallas; 24 Durham) were shown. Participants indicated the likelihood that they had actually visited that location (definitely yes, probably yes, might have, no). 24 non-home campus photos were new, and 48 were from Session 1 (24 once presented; 24 twice presented).

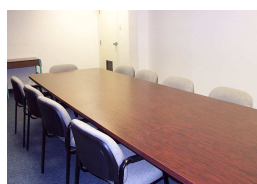
Sample Campus Photos

Mundane

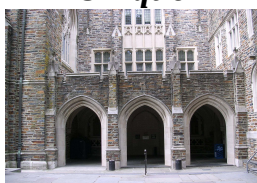
Duke



SMU



Unique

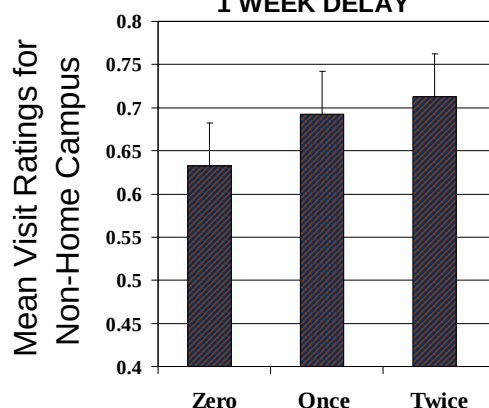


Experiment 1 Results

Results regardless of mundane and unique categories

Duke and SMU students

1 WEEK DELAY



Number of Times Photos Presented in First Session

Experiment 2 Method

Participants. 57 SMU students

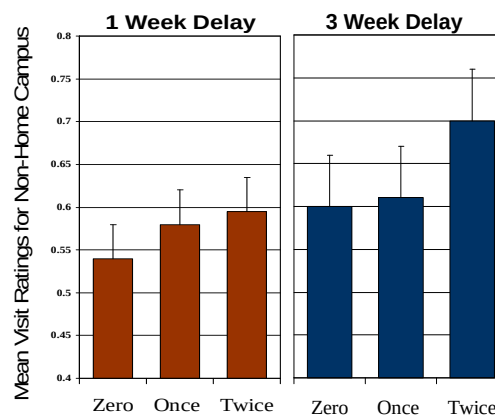
Purpose: Replicate Experiment 1, and compare 1 vs. 3 weeks.

Experiment 2 Results

Results regardless of mundane and unique categories

Only SMU Students

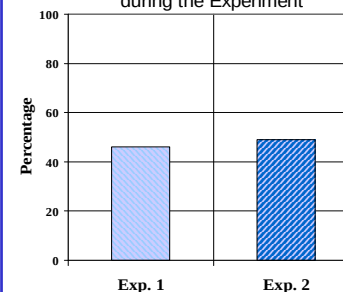
For both delay periods, increased exposure in the first session lead to a higher visit rating of the on-home campus.



Number of Times Photos Presented in First Session

Self Report Data: Déjà Vu Experiences

Percentage of Participants Reporting Déjà Vu Experiences during the Experiment



For both experiments, nearly half of the participants indicated feeling a sense of déjà vu during the experiment.

Conclusions

In two experiments, prior shallow processing of unfamiliar locations increased subjects' subsequent beliefs they had actually visited those places. Effects of prior exposure did not depend on the typicality of the unfamiliar locations, nor the delay between initial viewing and final test. Importantly, self-reports of subjective experiences supported the validity of the paradigm for studying déjà vu: about half of the participants in each experiment claimed to have experienced déjà vu or something approaching it during the experiment.

Our results fit with research on misattributions of implicit memories. With the *mere exposure effect*, subjects give higher liking ratings to previously flashed shapes even though they do not recognize them (Kunst-Wilson & Zajonc, 1980), and this effect is unaffected by a delay (Seamon, Brody, & Kauff, 1983). In the *false fame effect*, prior laboratory exposure to non-famous names is misinterpreted as pre-experimental fame after a delay (Jacoby, Kelley, Brown, & Jasechko, 1989). While some research on misattributions of familiarity or fluency have been explicitly linked to déjà vu (e.g., Jacoby & Whitehouse, 1989), our research goes beyond a list learning task to evaluate a false sense of having personally visited a real location.

References

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